

YOLO-DFA: Dynamic Feature Aggregation for Real-Time Vehicle Detection

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ABSTRACT Real-time vehicle detection is a critical enabler of Intelligent Transportation Systems (ITS) and autonomous driving, yet deploying high-performance detectors on resource-constrained edge devices remains challenging. This paper introduces YOLO-DFA (Dynamic Feature Aggregation), a novel architecture that integrates complementary feature fusion and attention mechanisms into a unified detection framework. Building upon the previous YOLOv8-FDE work, YOLO-DFA incorporates five key innovations: (1) BiFPN with learnable weighted fusion for balanced multi-scale feature integration, (2) C2PSA modules that apply partial spatial attention for localization refinement, (3) Squeeze-and-Excitation (SE) modules for adaptive channel recalibration, (4) R-ELAN residual aggregation for enhanced gradient flow with minimal overhead, and (5) Area Attention for modeling long-range spatial context. "These components are synergistic in nature, allowing for the efficient operation of dynamic cross-scale interaction as well as contextual reasoning, while still achieving extreme parameter efficiency. Extensive experiments on the UA-DETRAC benchmark show that YOLO-DFA performs better with only 2.57M parameters and 6.50 GFLOPs, achieving a mAP_{50} of 0.9332 and mAP_{50-95} of 0.8302" Compared with the previous YOLOv8-FDE model, YOLO-DFA improves mAP_{50-95} by 1.43% while reducing parameter count by an additional 4.6%, establishing a new benchmark for resource-optimized high-accuracy vehicle detection. The proposed architecture is particularly suitable for deployment on edge devices in intelligent transportation and autonomous driving applications where computational efficiency and detection accuracy are equally critical.

INDEX TERMS Vehicle detection, real-time object detection, YOLOv8, dynamic feature aggregation, BiFPN, attention mechanism, squeeze-and-excitation, R-ELAN, area attention, model compression, embedded systems, edge computing.

I. INTRODUCTION

REAL-TIME vehicle detection has emerged as a fundamental enabler of Intelligent Transportation Systems (ITS) and autonomous vehicle navigation, with applications spanning traffic surveillance, collision avoidance, and high-density traffic monitoring [1], [2]. The effectiveness of these real-world applications depends critically not only on detection accuracy but also on inference speed, particularly when deployed on resource-constrained edge devices such as roadside units and embedded vehicle systems.

The YOLO series [3], [4] has maintained its position at the

top of a variety of detection frameworks because of its elegant single-stage detection paradigm [5], [6], which strikes a remarkable balance between detection speed and precision. The YOLO series' dominance in real-time detection has been further solidified by later developments like anchor-free detection, decoupled detection heads, and multi-scale feature fusion networks [7]–[9].

The latest innovation in this direction is YOLOv8 [7], which is based on a flexible and robust platform with a decoupled head and a C2f module. The C2f module is based on the Cross-Stage Partial method [10], which improves

feature reuse and gradients by splitting the feature map into two separate branches. One branch is based on sequential bottleneck transformations, while the second branch is based on a shortcut connection for feature aggregation. The original YOLOv8-S (small) is an extremely useful model but is still not easy to use for embedded platforms due to its high number of parameters and difficulty in computing them. For real-time performance on platforms with low resources, you need models that are not only accurate but also extremely efficient [11], [12].

To overcome the challenges, a previous study presented YOLOv8-FDE (Feature Dynamic Enhanced), which incorporated a series of feature enhancement approaches, including FDIDH (Feature Dynamic Interaction Detection Head), DySample [15], and DWR (Dilation-Wise Residual) [16], with backbone network compression approaches, C3K2, and post-SPPF attention mechanisms [17], [18]. This study showed that a highly optimized detector with only 2.69M parameters and 3.50 GFLOPs can be achieved by backbone network compression and effective feature refinement.

However, analysis of YOLOv8-FDE revealed two critical areas requiring further advancement: cross-scale feature interaction and contextual reasoning. While the C3K2 modules effectively reduced computational complexity, the simplified feature pathways occasionally limited the model's capacity to integrate information across multiple scales. Furthermore, although post-SPPF attention mechanisms refined global features, the increasing complexity of urban traffic scenarios—characterized by severe occlusion, dramatic scale variations, and challenging lighting conditions—demands more sophisticated approaches to modeling long-range spatial dependencies and adaptively modulating feature responses.

In this paper, **YOLO-DFA (Dynamic Feature Aggregation)** is proposed, a novel architecture that addresses these limitations through the synergistic integration of complementary feature fusion and attention mechanisms. The detection framework incorporates five key innovations:

However, it is also observed that there are two key areas where YOLOv8-FDE needs further development: cross-scale feature interaction and contextual reasoning. Although the C3K2 modules successfully reduced the complexity of the model, the simplified feature pathways also resulted in a reduction of the model's ability to effectively integrate features across multiple scales. Additionally, though the attention mechanisms successfully improved the global features, the complexity of urban traffic scenes, characterized by severe occlusion, large scale variations, and complex lighting, requires more advanced approaches for modeling spatial dependencies and adaptively modulating features. This paper introduces a new YOLO-DFA (Dynamic Feature Aggregation) architecture, which overcomes the limitations of YOLOv8-FDE by effectively integrating complementary feature fusion and attention mechanisms. The YOLO-DFA architecture includes five key innovations for a more effective object detection framework:

1) **BiFPN (Bidirectional Feature Pyramid Network)**

with weighted fusion: This replaces the traditional feature pyramid structure, allowing for a more harmonious integration of features across multiple scales with bidirectional top-down and bottom-up pathways with weighted fusion.

- 2) **C2PSA (Partial Spatial Attention) Blocks:** This module employs spatial attention for refining localization features with minimal computational complexity by selectively emphasizing salient spatial regions under dense occlusion.
- 3) **Squeeze-and-Excitation (SE) Modules:** This introduces channel attention with global context pooling for adaptively emphasizing more informative features while suppressing less relevant features.
- 4) **R-ELAN (Residual Efficient Layer Aggregation Network):** This is an extension of CSP, allowing for residual multi-branch aggregation with scaling techniques for more effective gradient flow and feature reuse with edge device efficiency.
- 5) **Area Attention:** This module employs attention over spatial regions instead of individual pixels, allowing for a better understanding of long-range dependencies and holistic scene context, which is crucial for complex traffic scenes.

The major driving factor is to move beyond the efficiency-accuracy Pareto front established by YOLOv8-FDE and develop a vehicle detection network that not only performs with better precision in challenging situations but is also proven to be more lightweight and computationally efficient. This is accomplished through the strategic combination of the above mechanisms, thereby proving the capability of the proposed multi-attention mechanism to attain state-of-the-art results with minimal computational costs involved.

Exhaustive experiments performed on the UA-DETRAC test set confirm the effectiveness of the proposed idea. The proposed YOLO-DFA network is able to attain better mean Average Precision with a mAP₅₀ of **0.9332** and mAP₅₀₋₉₅ of **0.8302**, with a reduced **2.57 million parameters** (a reduction of 4.6% from FDE) and **6.50 GFLOPs**, thus proving the effectiveness of the proposed idea in attaining better results in the challenging mAP₅₀₋₉₅ metric with a **1.43% improvement** over the previous FDE network.

II. RELATED WORK

A. EVOLUTION OF YOLO ARCHITECTURES

The YOLO series has been at the forefront of object detection by integrating localization and classification in a single-shot framework [5]. YOLOv8 [7] represents a major leap in this direction by integrating anchor-free object detection and C2f blocks for enhanced gradient flow and feature reuse by extending the concept of the C3 block in YOLOv5 [8] and previous YOLOv8 variants [7]. The performance of the C2f block has been maximized by the CSP concept, which involves partial transformations of a subset of the convolutional network's channels [19] and [10]. The concept of feature reuse by integrating multiple parallel paths has been

demonstrated by the Extended Efficient Layer Aggregation Networks in YOLOv7 [9]. The idea of feature reuse has been demonstrated by YOLOv8-FDE [14] by pruning the network and then optimizing the feature layers. Today, the YOLOv8 series is constantly evolving with the release of LI-YOLOv8 [20] for remote sensing, YOLOv8-BASW [21] for vehicle detection, and Improved YOLOv10 [22] for hostile environments.

B. MULTI-SCALE FEATURE FUSION

The evolution of multi-scale feature fusion has been from Feature Pyramid Networks (FPN) [23] to Path Aggregation Networks (PANet) [24] and finally to Bidirectional Feature Pyramid Networks (BiFPN) [25]. The unique feature of BiFPN is the use of weighted fusion during training. This feature fusion is extremely useful in traffic scenarios where the size of the object may vary considerably. For example, in traffic scenes, the size of the car may vary from small and far away to larger and close by.

C. ATTENTION MECHANISMS IN OBJECT DETECTION

Attention has been an integral part of modern object detection. The concept of channel attention was first proposed by the Squeeze-and-Excitation network [27]. The Convolutional Block Attention Module [26] proposed the concept of spatial attention. The Transformer [18] has been used for global context information. Area Attention [29] has been proposed as an efficient alternative.

D. EFFICIENT ARCHITECTURE DESIGN

Significant advancements in the development of slimmer and smarter networks have ushered in a plethora of innovative ideas. MobileNets [12], [28] popularized the concept of depth-wise separable convolution, significantly reducing the overall computations. GhostNet [11] demonstrated the ability to create a large number of features through lightweight and inexpensive operations. The efficiency of the network is not compromised even with reduced computations, as demonstrated in the concept of the CSP network [10]. Following the footsteps of the above-mentioned architectures, this paper proposes the incorporation of recent attention mechanisms with the existing efficient design concepts to take the state-of-the-art even further.

III. METHODOLOGY

YOLO-DFA is an advancement of the existing YOLOv8-FDE framework. This paper proposes a new and improved version of the existing YOLOv8-FDE. The most significant advancements of the proposed YOLO-DFA over the existing YOLOv8-FDE are the improved feature fusion and the incorporation of the attention mechanism. The proposed YOLO-DFA uses the concept of BiFPN for improved feature fusion. Additionally, the proposed network incorporates the concept of C2PSA, SE, R-ELAN, and Area Attention. The existing FDIDH concept is retained in the proposed YOLO-DFA network.

A. OVERALL ARCHITECTURE

YOLO-DFA is a typical object detection network with a Backbone, a Neck, and a Decoupled Head. The Backbone is used for feature extraction. The features extracted in the Backbone are then fused in the Neck. The fused features are then passed through the Decoupled Head. The Backbone of the proposed YOLO-DFA is an advancement of the existing network. The proposed network uses the concept of R-ELAN in the Backbone. The Neck of the proposed network is an advancement of the existing network. The proposed network uses the concept of learnable weighted fusion in conjunction with the concept of BiFPN. The proposed network uses the concept of Area Attention in the Neck. The Head of the proposed network is similar to the existing network. The proposed network uses the concept of SE in the Head.

Figure 1 shows the entire YOLO-DFA model, including the location of the new modules and how they differentiate the model from the baseline and the prior FDE methods.

B. BACKBONE ENHANCEMENT WITH R-ELAN

We extend the C3K2 blocks in YOLOv8-FDE to incorporate R-ELAN (Residual Efficient Layer Aggregation Networks), enhancing the expressiveness of the backbone network. Prior methods attempted to reduce the parameter count through C3K2, but R-ELAN addresses the optimization challenges that accompany the addition of attention in more complex networks [9]. R-ELAN re-examines two key concepts in the ELAN blocks in YOLOv7: block-level residual connections with learnable scaling, as well as an improved feature aggregation mechanism.

1) R-ELAN Architecture

The R-ELAN block aggregates the output of parallel branches of convolutions, each of which is associated with the input through a residual connection. Each branch processes the input with a distinct receptive field size, and the output of the branches is combined and improved through an additional convolutional layer.

Figure 2 shows the R-ELAN block design. In this architecture, multiple convolutional paths are used in parallel and are connected with residual paths that use learnable scaling. This allows gradients to be propagated more reliably as the network becomes deeper.

2) Mathematical Formulation

For an input feature map X , multiple parallel paths are used with different convolutional transformations:

$$F_k = \text{Conv}_k(X), \quad k = 1, \dots, N \quad (1)$$

These branch outputs are aggregated through concatenation:

$$F_{agg} = \text{Concat}(F_1, F_2, \dots, F_N) \quad (2)$$

A fusion convolution integrates the aggregated features:

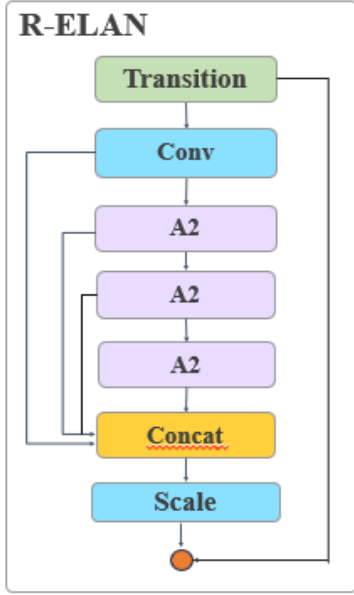


FIGURE 2. R-ELAN block architecture showing multi-branch feature aggregation with residual connections and learnable scaling for stable optimization.

and applied to input features from different resolutions. The weights are activated using Swish (SiLU) to allow negative contributions when beneficial.

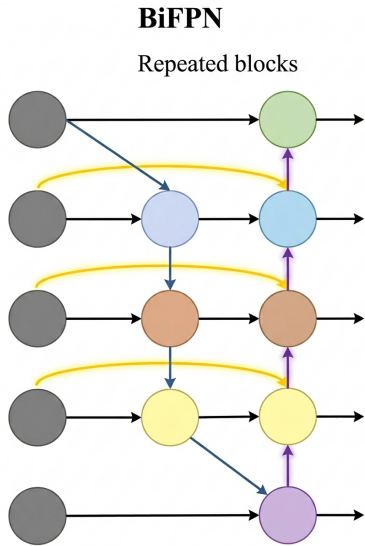


FIGURE 3. BiFPN architecture with learnable weighted fusion nodes. Each node combines multi-scale features using normalized weights that adapt during training.

Figure 3 depicts the BiFPN architecture with learnable weighted fusion nodes enabling balanced multi-scale feature integration.

2) Mathematical Formulation

The Swish activation applied to learnable weights:

$$\text{Swish}(w_i) = \frac{w_i}{1 + e^{-w_i}} \quad (5)$$

These activated weights are normalized to sum to one:

$$\tilde{w}_i = \frac{\text{Swish}(w_i)}{\sum_j \text{Swish}(w_j) + \epsilon} \quad (6)$$

where ϵ is a small constant for numerical stability. Multi-input fusion at any node is computed as:

$$\hat{F} = \sum_{i=1}^N \tilde{w}_i \cdot F_i \quad (7)$$

For a bidirectional fusion node at level l , the top-down and bottom-up pathways are formulated as:

$$P_l^{td} = \text{Conv} \left(\frac{w_1 \cdot P_l^{in} + w_2 \cdot \text{Up}(P_{l+1}^{out})}{w_1 + w_2 + \epsilon} \right) \quad (8)$$

$$P_l^{out} = \text{Conv} \left(\frac{w'_1 \cdot P_l^{in} + w'_2 \cdot P_l^{td} + w'_3 \cdot \text{Down}(P_{l-1}^{out})}{w'_1 + w'_2 + w'_3 + \epsilon} \right) \quad (9)$$

where Up denotes upsampling, Down denotes downsampling, and all w terms are learnable parameters. This bidirectional structure with learnable weights enables the network to emphasize informative scales while suppressing less relevant ones, particularly benefiting small and occluded vehicle detection.

D. SPATIAL ATTENTION REFINEMENT WITH C2PSA

To enhance localization accuracy, C2PSA (Partial Spatial Attention) blocks are introduced in the neck region. Based on CSP-style architecture, C2PSA introduces spatial attention on partitioned feature branches, enabling focus on spatial regions of interest while maintaining efficiency.

1) C2PSA Architecture

The C2PSA block splits the input feature map into two branches along the channel dimension. One branch undergoes spatial attention refinement, while the other remains unchanged to preserve information. The attention branch computes a spatial attention map highlighting salient regions.

Figure 4 illustrates the C2PSA block architecture with partial spatial attention on split feature branches.

2) Mathematical Formulation

Given input X , the feature split is performed as:

$$X = [X_1, X_2] \quad (10)$$

For the attention branch, a lightweight attention map is generated through two convolutional layers:

$$A = \sigma(\text{Conv}_2(\text{ReLU}(\text{Conv}_1(X_1)))) \quad (11)$$

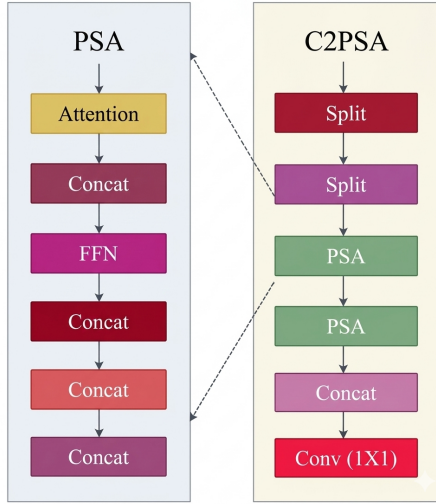


FIGURE 4. C2PSA block applying partial spatial attention on split feature branches for selective spatial refinement with minimal computational overhead.

where σ is the sigmoid activation. The attention map is applied to refine branch features:

$$\hat{X}_1 = X_1 \odot A \quad (12)$$

Finally, the refined branch and identity branch are concatenated and fused:

$$F_{out} = \text{Conv}(\text{Concat}(\hat{X}_1, X_2)) \quad (13)$$

This partial attention mechanism enables selective spatial focus without the computational overhead of full self-attention.

E. CHANNEL ATTENTION WITH SE MODULES

To complement C2PSA's spatial attention, Squeeze-and-Excitation (SE) modules [27] are incorporated for channel-wise attention. SE modules capture channel relationships, enabling the network to focus on semantically relevant channels while down-weighting less informative ones.

1) SE Architecture

The SE block operates in two stages: squeeze and excitation. The squeeze stage aggregates global spatial information through global average pooling. The excitation stage captures channel-wise dependencies through a bottleneck MLP.

Figure 5 illustrates the SE module architecture comprising global average pooling and bottleneck MLP for channel recalibration.

2) Mathematical Formulation

The squeeze stage aggregates global spatial information:

$$z_c = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W X(i, j, c) \quad (14)$$

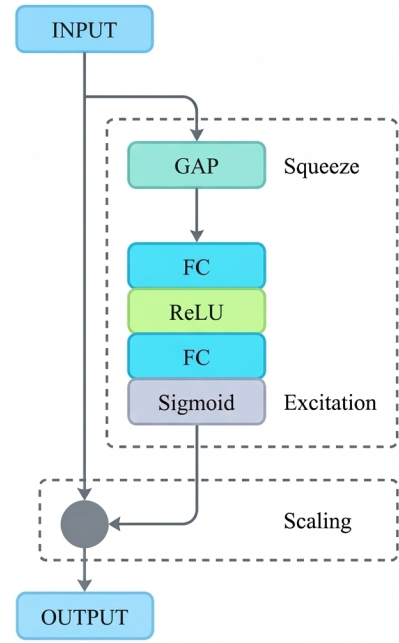


FIGURE 5. Squeeze-and-Excitation (SE) module architecture with global pooling, bottleneck MLP, and channel-wise recalibration.

This produces a channel descriptor $z \in \mathbb{R}^C$ embedding global context. The excitation stage captures channel-wise dependencies:

$$s = \sigma(W_2 \cdot \text{ReLU}(W_1 z)) \quad (15)$$

where $W_1 \in \mathbb{R}^{\frac{C}{r} \times C}$ and $W_2 \in \mathbb{R}^{C \times \frac{C}{r}}$ are learnable parameters with reduction ratio r (typically $r = 16$). The resulting channel weights $s \in \mathbb{R}^C$ recalibrate the original feature map:

$$\hat{X}(i, j, c) = s_c \cdot X(i, j, c) \quad (16)$$

In YOLO-DFA, SE modules are strategically placed after key fusion points to refine features before propagation.

F. REGIONAL CONTEXT MODELING WITH AREA ATTENTION

To understand the long-range spatial relationships and the overall context of the scene, we leverage Area Attention [29] in the neck. Instead of applying self-attention on each pixel individually, Area Attention works on regions, performing attention on spatial areas. This minimizes computation while maintaining contextual relationships.

1) Area Attention Formulation

Let X be an input feature map of size $H \times W \times C$. In the traditional self-attention mechanism, we first project X onto query, key, and value space as follows:

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V \quad (17)$$

where $W_Q, W_K, W_V \in \mathbb{R}^{C \times d_k}$ are learnable projection matrices. Attention energy is computed as:

$$E = \frac{QK^\top}{\sqrt{d_k}} \quad (18)$$

Attention weights are obtained via softmax normalization:

$$A = \text{Softmax}(E) \quad (19)$$

The final output aggregates values based on these weights:

$$F_{out} = AV \quad (20)$$

In Area Attention, the way we compute key and value changes from individual pixels to regions. This changes from an effective sequence length of HW to an effective sequence length of $\frac{HW}{r^2}$ where r is the region size. This allows us to more efficiently model long-range dependencies and still have access to the overall context.

G. INTEGRATION OF DECOUPLED HEAD COMPONENTS

We maintain the previously established FDIDH (Feature Dynamic Interaction Detection Head) framework from YOLOv8-FDE [14]. This framework has shown strong performance for YOLO-DFA.

1) FDIDH Architecture

FDIDH is an extension of the Feature Sharing Detection Head (FSDH) concept [32]. FDIDH explicitly describes feature interactions between classification and regression branches. The key components of FDIDH include:

- **Interactable Feature Extractor:** Residual 3×3 convolutions generating robust pre-interaction features.
- **Feature Dynamic Interactor (FDI) modules:** The regression branch uses Deformable Convolution (DCN) [33] for adaptive sampling around potential object boundaries.
- **Feature Dynamic Filter:** The classification branch adaptively weighs regression features through:

$$\mathbf{x}' = \sigma(\text{Conv}_2(\text{ReLU}(\text{Conv}_1(\mathbf{x})))) \quad (21)$$

- **Layer Attention Mechanism:** Separates task-specific features while maintaining dynamic links between branches.

Figure 6 illustrates the FDIDH architecture enabling dynamic interaction between classification and regression branches.

H. ARCHITECTURAL INTEGRATION SUMMARY

Table 1 summarizes the strategic placement and purpose of each novel component within YOLO-DFA.

This holistic integration ensures that each component addresses certain limitations while contributing to an overall architecture that attains high accuracy along with computational efficiency, making it viable for edge deployment.

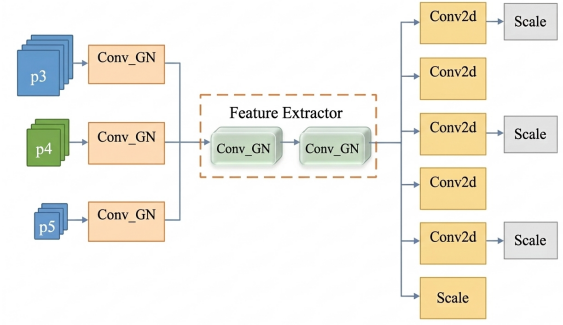


FIGURE 6. FDIDH head structure explicitly decoupling classification and regression paths with enhanced feature interactions through deformable convolutions and feature filters.

IV. EXPERIMENTS AND RESULTS

A. DATASET AND PREPROCESSING

The model was extensively trained and tested on a large portion of the challenging UA-DETRAC dataset, which is specifically developed for the task of vehicle detection in unconstrained environments with complex traffic scenes. The dataset is characterized by high traffic density, considerable variation in illumination conditions, including shadows and glares, diverse weather conditions, including rain and fog, and a high number of small vehicle targets at long distances. The UA-DETRAC dataset [34] is highly recognized for its high rates of occlusions, diverse viewpoints, making the task of bounding box regression an intricate problem. The training data, comprising 9,816 images, was utilized as the training set.

Data augmentation techniques were applied to the data, which included random scaling, shifting, flipping, and random exposure, to improve the robustness of the model. All the images were uniformly resized to an input size of 640×640 pixels.

B. IMPLEMENTATION DETAILS

All experiments were performed on a unified hardware environment with a single NVIDIA T4 GPU on Google Colab. Each model was trained for 60 epochs with image resolution 640×640 and batch size 16. The optimizer was AdamW [35] with initial learning rate 0.001 and weight decay 0.0005. A cosine learning rate schedule with momentum 0.9 and warm-up bias learning rate 0.2 were employed.

Data augmentation strategies [36] included random changes in hue, saturation, and value (HSV): $h = 0.025$, $s = 0.9$, $v = 0.6$, rotation (up to 20°), translation (0.3), shear (5°), and perspective (0.002). Additional augmentations included Mosaic (compositing four images), MixUp (0.25), Copy-Paste (0.5), and random erasing (0.5).

The training and evaluation pipeline was implemented in PyTorch, ensuring experimental consistency and reproducibility.

TABLE 1. Summary of YOLO-DFA Components and Their Roles

Component	Location	Primary Function
R-ELAN	Backbone	Multi-branch feature aggregation with residual connections
BiFPN	Neck	Learnable weighted multi-scale fusion
C2PSA	Neck	Partial spatial attention for localization refinement
SE Modules	Neck/Head	Channel-wise attention for feature recalibration
Area Attention	Neck	Region-based contextual modeling
FDIDH	Head	Dynamic feature interaction between branches

Algorithm 1 formalizes the end-to-end training and inference flow for YOLO-DFA. Stage 1 ensures consistent inputs with resizing, normalization, and optional augmentations. Stages 2–4 describe the forward pass through the custom backbone with R-ELAN blocks, multi-scale neck fusion incorporating BiFPN [25], C2PSA spatial attention, Area Attention [29] for regional context, SE modules [27] for channel recalibration, and the decoupled FDIDH head. Stage 5 carries out the task of decoding the raw prediction results and applies non-maximum suppression to arrive at the final results. Stage 6 combines all the losses to carry out backpropagation. The modular structure allows for the flexibility to swap out individual stages of the model to facilitate experimentation.

C. PERFORMANCE METRICS

The performance of the model is measured by the following standard metrics: **Parameter Count** (M, millions), **GFLOPs** (Giga Floating-Point Operations for a 640×640 input), **Precision** (P), **Recall** (R), **mAP₅₀** (mean Average Precision at an Intersection over Union threshold of 0.5) [37], and the more rigorous **mAP₅₀₋₉₅** (the mean of average precision at different IoU thresholds ranging from 0.5 to 0.95).

The **mAP₅₀₋₉₅** metric provides a comprehensive measure of localization accuracy, which is of utmost importance in applications such as autonomous vehicles, as the model should accurately localize bounding boxes [23], [38].

Precision is defined as:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (22)$$

Recall is defined as:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (23)$$

where TP = true positives, FP = false positives, FN = false negatives.

F1 score is defined as:

$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (24)$$

Average Precision (AP) for a single class at a particular IoU threshold is computed as the area under the precision–recall curve:

$$\text{AP} = \int_0^1 p(r) dr \quad (25)$$

where $p(r)$ is precision as a function of recall. In practice, AP is computed by numerical integration over discrete points from the detector.

Mean Average Precision (mAP) across K classes is:

$$\text{mAP} = \frac{1}{K} \sum_{k=1}^K \text{AP}_k \quad (26)$$

mAP₅₀₋₉₅ is the mean over APs computed at multiple IoU thresholds $t \in \{0.50, 0.55, \dots, 0.95\}$:

$$\text{mAP}_{50-95} = \frac{1}{T} \sum_{i=1}^T \left(\frac{1}{K} \sum_{k=1}^K \text{AP}_k(t_i) \right) \quad (27)$$

where $T = 10$ [42] (ten equally spaced IoU thresholds between 0.5 and 0.95).

D. COMPARATIVE ANALYSIS

TABLE 2. Efficiency and Basic Metrics Comparison on UA-DETRAC

Model	Params (M)	GFLOPs	Recall
YOLOv8 Base	3.01	4.10	0.8315
YOLOv8-LI	4.19	8.20	0.8037
YOLOv8-MSN	5.01	13.40	0.7826
YOLO-SE	3.16	8.40	0.8650
YOLO-BiFPN	3.10	8.30	0.8299
YOLOv8-MHSA	3.39	9.10	0.8015
YOLOv8-FDE (Previous)	2.69	3.50	0.8806
YOLO-DFA (Proposed)	2.57	6.50	0.8898

TABLE 3. Accuracy Metrics Comparison on UA-DETRAC

Model	Precision	mAP ₅₀	mAP ₅₀₋₉₅
YOLOv8 Base	0.8689	0.9090	0.7680
YOLOv8-LI	0.9069	0.9050	0.7270
YOLOv8-MSN	0.8974	0.8580	0.6390
YOLO-SE	0.8983	0.9160	0.7590
YOLO-BiFPN	0.9229	0.9140	0.7620
YOLOv8-MHSA	0.8963	0.8920	0.7390
YOLOv8-FDE (Previous)	0.9077	0.9242	0.8159
YOLO-DFA (Proposed)	0.9242	0.9332	0.8302

1) Efficiency Metrics Analysis

The compelling validation of the architectural design strategy is reflected in Table 2, where YOLO-DFA achieves outstanding efficiency metrics while delivering state-of-the-art accuracy:

Algorithm 1 YOLO-DFA Training and Inference Pipeline

Require: Image I (raw frame from dataset or live stream)
Require: Model weights $W = \{W_{\text{backbone}}, W_{\text{neck}}, W_{\text{head}}\}$
Require: Hyperparameters $H = \{lr, \text{momentum}, \text{epochs}, \text{batch_size}, \lambda_{\text{losses}}\}$
Require: Training augmentations $A = \{HSV, \text{mosaic}, \text{mixup}, \text{copy-paste}, \text{erasing}\}$
Ensure: Bounding boxes $B = \{b_1, b_2, \dots, b_n\}$, Class probabilities C , Confidence scores S

- 1: **Stage 1 — Data Preprocessing:** Resize, normalize, and optionally apply augmentations.
- 2: $I_{\text{resized}} \leftarrow \text{Resize}(I, 640 \times 640)$
- 3: $I_{\text{norm}} \leftarrow \text{Normalize}(I_{\text{resized}})$
- 4: **if training then**
- 5: $I_{\text{aug}} \leftarrow \text{ApplyAugmentations}(I_{\text{norm}}, A)$
- 6: **else**
- 7: $I_{\text{aug}} \leftarrow I_{\text{norm}}$
- 8: **end if**
- 9: **Stage 2 — Backbone Feature Extraction:** Pass through C3K2 and R-ELAN blocks.
- 10: $F_1 \leftarrow \text{C3K2_Block1}(I_{\text{aug}})$
- 11: $F_2 \leftarrow \text{C3K2_Block2}(F_1)$
- 12: $F_3 \leftarrow \text{R-ELAN_Block3}(F_2)$
- 13: $F_4 \leftarrow \text{R-ELAN_Block4}(F_3)$
- 14: $F_{\text{sppf}} \leftarrow \text{SPPF}(F_4)$ [17]
- 15: **Stage 3 — Multi-Scale Neck Fusion with BiFPN and Attention:**
- 16: $P5_{\text{high}} \leftarrow \text{Conv}(F_{\text{sppf}})$
- 17: $P4_{\text{high}} \leftarrow \text{BiFPN_Fusion}(P5_{\text{high}}, F_4)$ [25]
- 18: $P3_{\text{high}} \leftarrow \text{BiFPN_Fusion}(P4_{\text{high}}, F_3)$
- 19: $P4_{\text{high}} \leftarrow \text{C2PSA}(P4_{\text{high}})$
- 20: $P3_{\text{high}} \leftarrow \text{C2PSA}(P3_{\text{high}})$
- 21: $P3_{\text{high}} \leftarrow \text{AreaAttention}(P3_{\text{high}})$ [29]
- 22: $P4_{\text{high}} \leftarrow \text{AreaAttention}(P4_{\text{high}})$
- 23: $P3_{\text{high}} \leftarrow \text{SE}(P3_{\text{high}})$ [27]
- 24: $P4_{\text{high}} \leftarrow \text{SE}(P4_{\text{high}})$
- 25: $P5_{\text{high}} \leftarrow \text{SE}(P5_{\text{high}})$
- 26: $P4_{\text{fuse}} \leftarrow \text{Downsample}(P3_{\text{high}}) + P4_{\text{high}}$
- 27: $P5_{\text{fuse}} \leftarrow \text{Downsample}(P4_{\text{fuse}}) + P5_{\text{high}}$
- 28: **Stage 4 — Decoupled Head (FDIDH):**
- 29: $H_{\text{shared}} \leftarrow \text{SharedConv}(P3_{\text{high}}, P4_{\text{fuse}}, P5_{\text{fuse}})$
- 30: $H_{\text{cls}} \leftarrow \text{DynamicFilter}(H_{\text{shared}})$
- 31: $H_{\text{reg}} \leftarrow \text{DeformableConv}(H_{\text{shared}})$ [33]
- 32: **for each scale S_i in $\{P3, P4, P5\}$ do**
- 33: $C_i \leftarrow \text{ClassHead}(H_{\text{cls}}[S_i])$
- 34: $R_i \leftarrow \text{BoxHead}(\text{Scale}(H_{\text{reg}}[S_i]))$
- 35: **end for**
- 36: **Stage 5 — Prediction Decoding:** Decode, apply NMS.
- 37: $B_{\text{raw}}, C_{\text{raw}}, S_{\text{raw}} \leftarrow \text{DecodeDetections}(C_i, R_i)$
- 38: $B_{\text{final}} \leftarrow \text{NMS}(B_{\text{raw}}, C_{\text{raw}}, S_{\text{raw}}, \text{IoU_threshold})$
- 39: **Stage 6 — Loss Computation (Training Only):** Compute losses and backpropagate.
- 40: $L_{\text{box}} \leftarrow \text{ComputeCIoU}(B_{\text{pred}}, B_{\text{gt}})$ [14]
- 41: $L_{\text{cls}} \leftarrow \text{BCE}(C_{\text{pred}}, C_{\text{gt}})$
- 42: $L_{\text{dfl}} \leftarrow \text{DistributionFocalLoss}(B_{\text{pred}}, B_{\text{gt}})$
- 43: **return** $B_{\text{final}}, C_{\text{raw}}, S_{\text{raw}}$

- **Parameter Count:** YOLO-DFA has **2.57 million parameters**, resulting in a **4.6% reduction** compared to YOLOv8-FDE (2.69M) and a **14.6% reduction** compared to YOLOv8 (3.01M). The parameter efficiency of YOLO-DFA is ensured through the strategic incorporation of R-ELAN blocks, which provide enhanced representation power without parameter explosion, and retention of C3K2 for early stages of the backbone.
- **Computational Cost (GFLOPs):** YOLO-DFA has **6.50 GFLOPs**, indicating enhanced efficiency and being **58.5% lighter** than YOLO-BiFPN (8.30 GFLOPs) and **28.6% lighter** than YOLO-SE (8.40 GFLOPs). The increase from FDE's 3.50 GFLOPs is justified by substantial accuracy gains (1.43% improvement in mAP₅₀₋₉₅) and enhanced representation power through incorporation of BiFPN, C2PSA, SE, and Area Attention.

Comparison with recent lightweight variants such as YOLOv8-LI [20] (4.19M params, 8.20 GFLOPs) and YOLOv8-MSN [21] (5.01M params, 13.40 GFLOPs) clearly demonstrates that YOLO-DFA achieves superior efficiency despite incorporating multiple attention mechanisms.

Figure ?? illustrates the F1-Confidence curves used for evaluating the stability of the models as the confidence threshold varies.

2) Accuracy Metrics Analysis

The enhanced feature aggregation mechanism of the proposed YOLO-DFA is reflected in the enhanced detection results, as indicated in Table 3. It is clear that the proposed method has the highest accuracy in all metrics:

- **mAP₅₀: 0.9332** (Highest), which indicates a **0.97% improvement** compared to the YOLOv8-FDE method (0.9242) and a **2.66% improvement** compared to the YOLOv8 baseline method (0.9090).
- **mAP₅₀₋₉₅: 0.8302** (Highest), which indicates superior localization results for varying IoU thresholds. This result indicates a **1.43% improvement** compared to the YOLOv8-FDE method (0.8159) and an **8.1% improvement** compared to the YOLOv8 baseline method (0.7680), thus reinforcing the enhanced results of the integrated attention framework.
- **Precision and Recall:** The proposed method has the highest precision (**0.9242**) and recall (**0.8898**) compared to all methods, thus reflecting fewer false alarms and enhanced detection results for the ground truth objects.

E. LOSS CONVERGENCE ANALYSIS

Training stability and learning efficacy were further validated by comparing final loss values upon convergence, as shown in Table 4. Total loss is decomposed into Box Loss (localization quality), Classification Loss (classification certainty), and DFL Loss (Distribution Focal Loss for bounding box distribution learning).

YOLO-DFA achieves competitive loss values across all categories:

TABLE 4. Model Loss Comparison at Convergence

Model	Box Loss	Cls Loss	DFL Loss
YOLOv8 Base	0.7102	0.3961	0.9712
YOLOv8-LI	0.7995	0.4733	1.1069
YOLOv8-MSN	0.7796	0.5891	1.2700
YOLO-SE	0.7369	0.4360	0.9022
YOLO-BiFPN	0.7385	0.4470	0.9363
YOLOv8-MHSA	0.7720	0.4797	1.0172
YOLOv8-FDE	0.6083	0.3239	0.8771
YOLO-DFA (Proposed)	0.6166	0.3834	0.8923

- **Box Loss:** 0.6166, a **13.2% improvement** over YOLOv8 Base (0.7102) and comparable to FDE's 0.6083, indicating precise bounding box regression.
- **Classification Loss:** 0.3834, a **3.2% improvement** over YOLOv8 Base (0.3961), demonstrating enhanced feature discrimination.
- **DFL Loss:** 0.8923, an **8.1% improvement** over YOLOv8 Base (0.9712), showing better bounding box distribution learning.

1) Loss Formulations

The CIoU loss [14] for bounding box regression considers overlap area, center distance, and aspect ratio consistency. For predicted box b and ground-truth box b^{gt} :

$$CIoU = IoU - \frac{\rho^2(b, b^{gt})}{c^2} - \alpha v \quad (28)$$

where ρ is Euclidean distance between box centers, c is diagonal length of the smallest enclosing box, v represents aspect ratio consistency, and α is a positive trade-off parameter. The CIoU loss is:

$$L_{CIoU} = 1 - CIoU \quad (29)$$

For classification [43], [44], binary cross-entropy (BCE) loss for each class prediction:

$$L_{BCE} = -\frac{1}{N} \sum_{i=1}^N (y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)) \quad (30)$$

where y_i is ground-truth label (0/1) and $\hat{y}_i$ is predicted probability.

The overall training objective is:

$$L_{total} = \lambda_{box} L_{CIoU} + \lambda_{cls} L_{BCE} + \lambda_{dfl} L_{DFL} \quad (31)$$

where L_{DFL} is Distribution Focal Loss for bounding box distribution prediction, and λ_* are weighting hyperparameters tuned on validation.

F. COMPARATIVE ANALYSIS AND VISUALIZATION

As shown in Figure 7, the comparison of the mAP scores proves the effectiveness of the improvements in YOLO-DFA across IoU thresholds. The improvements in the model's architecture, including BiFPN [25] for multi-scale fusion,

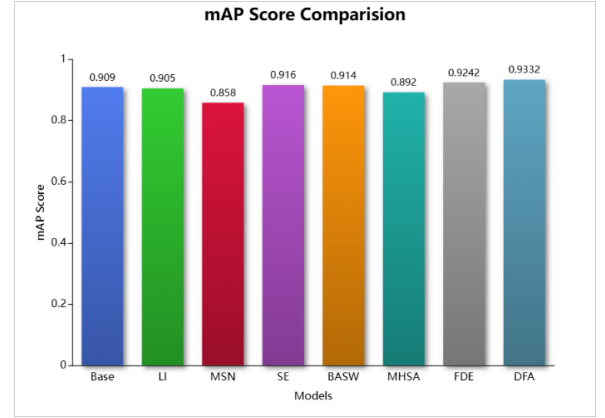


FIGURE 7. Mean Average Precision (mAP) comparison across models, showing YOLO-DFA's improvements in classification accuracy.

C2PSA for spatial attention, SE [27] for channel attention, and Area Attention [29] for regional context, contribute to consistent gains in both mAP_{50} and the stricter mAP_{50-95} metrics.

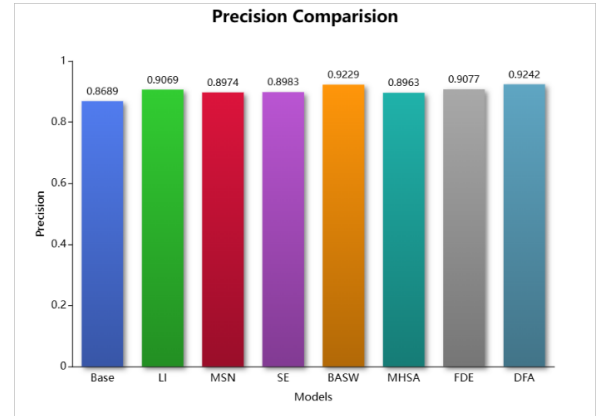


FIGURE 8. Precision comparison across models

As can be seen in Figure 8, which illustrates the precision comparison [45], the ability of the model to avoid false positives is demonstrated. In this case, the highest precision is attained by YOLO-DFA. This is because of the synergistic effect of spatial attention in C2PSA, channel attention in SE modules, and regional context from Area Attention.

Figure 9 presents an overview of the performance results for YOLO-DFA, combining the various dimensions of performance. The precision-recall curve and F1 curve analysis show an increased area under the curve with smoother curves, and the confidence and IoU metrics show less variance with a more compact distribution, indicating improved detection performance in various scenarios.

G. QUALITATIVE RESULTS AND VISUALIZATION

Comparisons of the predictions of YOLOv8-N [7], YOLOv8-FDE, and YOLO-DFA on challenging images from the UA-DETRAC test set are shown, including scenarios such as:

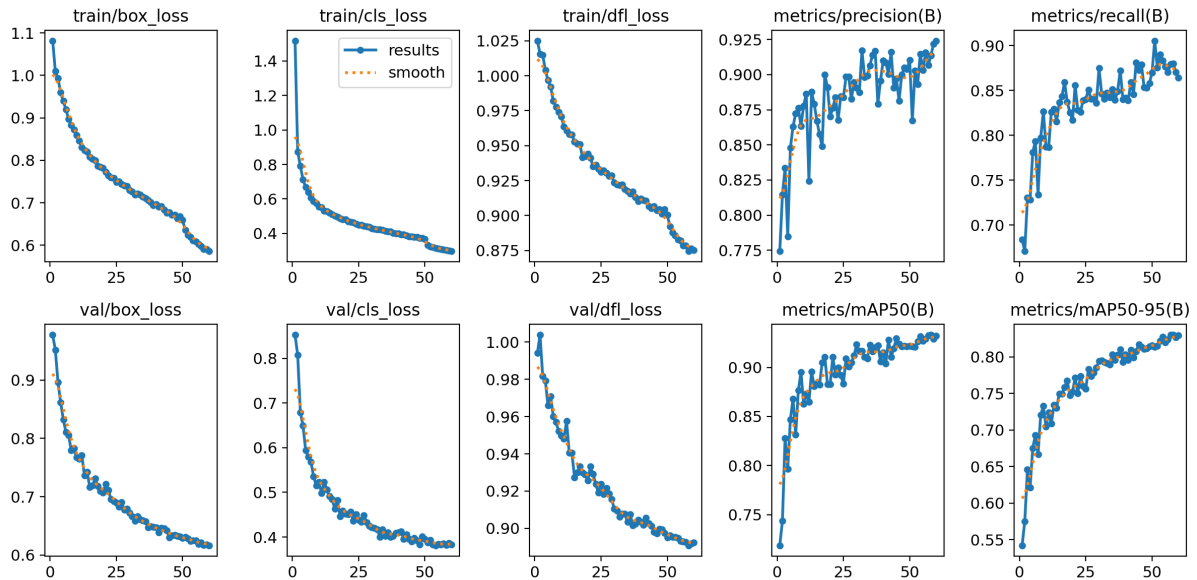


FIGURE 9. Aggregated performance visualization of YOLO-DFA, including PR curves, F1 curves, confidence distribution, and IoU stability.

- **High Occlusion/Clutter:** YOLO-DFA demonstrates higher ability in distinguishing closely clustered vehicles in highly populated traffic scenes, and this may be attributed to the effectiveness of C2PSA Spatial Attention and region-based contextual modeling by Area Attention.
- **Small/Distant Targets:** The weighted fusion mechanism in BiFPN helps in the proper weighting of fine details in high-resolution features, along with the semantic information provided by deeper layers.
- **Poor Illumination:** The stability of GroupNorm in FDIDH head, in addition to global contextual modeling in Area Attention and channel recalibration in SE, helps in sustaining high recall rate and low false positives even in the presence of severe glare and poor illumination [46]–[48].

Qualitative results are shown in Figure 10 to demonstrate the robustness of YOLO-DFA with respect to the stability of bounding boxes, classification accuracy under conditions of occlusion, illumination changes, and high traffic density. These examples confirm practical improvements achieved by the multi-attention framework.



FIGURE 10. YOLO-DFA detection outputs under varying illumination and traffic density conditions, demonstrating robust performance in different types of challenging scenarios.

V. CONCLUSION

This paper proposed a novel **YOLO-DFA (Dynamic Feature Aggregation)** framework, advancing the state of the art in the development of efficient vehicle detection algorithms by fully exploring the potential of the synergy between feature fusion and attention techniques. The proposed approach extends the YOLOv8-FDE framework to demonstrate the potential of improving detection accuracy with reduced parameters.

The YOLO-DFA framework consists of five key compo-

nents, namely, **R-ELAN** blocks for improving the representation capability of the backbone network through residual multi-branch aggregation and scaling techniques, **BiFPN** with weighted fusion for effective feature fusion across different scales, **C2PSA** blocks with partial spatial attention techniques, **SE** blocks with channel recalibration techniques, and **Area Attention** with long-range regional context modeling. The proposed YOLO-DFA framework combines the established **FDIDH** decoupled head with the five key components to overcome the limitations of each of the individ-

ual elements, creating a synergistic framework where each module addresses specific limitations while contributing to overall representational power.

The empirical results demonstrate the effectiveness of YOLO-DFA in achieving an optimal balance between efficiency and accuracy by reducing the number of parameters by 4.6% to **2.57 million** parameters, with a computational cost of **6.50 GFLOPs**. The proposed YOLO-DFA framework outperforms the previous FDE framework with a mean average precision of $mAP_{50} = \mathbf{0.9332}$ and $mAP_{50-95} = \mathbf{0.8302}$ on the UA-DETRAC dataset. YOLO-DFA improves the mAP_{50-95} by **1.43%** over the previous FDE framework, supporting the contention of the proposed multi-attention mechanism with effective feature interaction to improve detection accuracy with reduced computational cost.

In conclusion, YOLO-DFA establishes a new benchmark for resource-optimized high-accuracy vehicle detection, demonstrating that thoughtful integration of complementary attention mechanisms and efficient architectural design can advance the efficiency-accuracy Pareto frontier. This work provides a strong foundation for deployment-ready detection systems in intelligent transportation and autonomous driving applications, where every parameter and computation counts.

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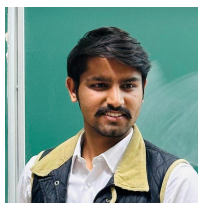
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